

Set Theory (Quite Naive):

Basic sets (whose existence we take for granted),

$$\mathbb{N} = \{1, 2, \dots\} \quad \mathbb{Z} = \{\dots, -1, -2, 0, 1, 2, \dots\}, \quad \mathbb{Q} = \left\{ \frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N} \right\}.$$

Inclusion Given sets A, B

$$A \subseteq B \text{ if } x \in A \Rightarrow x \in B, \text{ or } A \subset B$$

For emphasis, $A \subset B$ if $A \subseteq B$ and $A \neq B$.

$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A.$$

Power Set: $\mathcal{P}(X) = \{A : A \subseteq X\}$, where X is a set.

$\emptyset \in \mathcal{P}(X)$, i.e., $\emptyset$ is always a subset of X .

Union and Intersections: $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X) \leadsto \text{universe}$

$$A \cup B = \{x \in X, x \in A \text{ or } x \in B\}, \quad A \cap B = \{x \in X, x \in A \text{ and } x \in B\}$$

$$\bigcup_{i \in I} A_i = \{x \in X : x \in A_i \text{ for some } i \in I\}$$

$$\bigcap_{i \in I} A_i = \{x \in X : x \in A_i \text{ for every } i \in I\}$$

$$\bigcup_{i \in \emptyset} A_i = \emptyset$$

$$\bigcap_{i \in \emptyset} A_i = X : x \in \bigcap_{i \in \emptyset} A_i \Leftrightarrow x \in A_i, \text{ for all } i \in \emptyset \\ \Leftrightarrow x \in X$$

Difference Sets $A, B \in \mathcal{P}(X)$.

$$A \setminus B = \{x \in X : x \in A \text{ and } x \notin B\}$$

$X \setminus B$ is the complement of B .

Note : $A \setminus B = A \cap (X \setminus B)$

De Morgan's Laws: $\{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$

$$X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i), \quad X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$$

Product Sets: X, Y two sets:

$$X \times Y = \{(x, y), x \in X, y \in Y\}$$

$$\text{collection } \{X_i\}_{i \in I} \text{ of sets: } \prod_{i \in I} X_i = \{(x_i)_{i \in I} : x_i \in X_i\}$$

Axiom of Choice If $\{X_i\}_{i \in I}$ is an infinite collection of non-empty sets,

$$\text{then } \prod_{i \in I} X_i \neq \emptyset.$$

Let X, Y be non-empty sets. A function is a subset $f \subseteq X \times Y$

s.t. for each x in X , there is exactly one $y \in Y$ s.t. $(x, y) \in f$.

$$\text{Hence, write } y = f(x), \text{ so } f = \{(x, f(x)), x \in X\} \subseteq X \times Y$$

we think of $f: X \rightarrow Y$ as an operation.

Now let I be a non-empty set. Let $X_i = X$ (X is fixed nonempty set)

$$\text{Then } X^I = \prod_{i \in I} X_i = \{(x_i)_{i \in I}, x_i \in X, \text{ for each } i \text{ in } I\}.$$

$$X^I \ni (x_i)_{i \in I} \rightsquigarrow \begin{matrix} f: I \rightarrow X \\ f(i) = x_i \end{matrix}$$

$$X^I = \{f: I \rightarrow X\} \text{ (set of functions)}$$

$\Rightarrow X^I \neq \emptyset$, since there is an identity function in X^I .

Prop'n: Axiom of choice (AC) is equivalent to (AC'):

Given any $X \neq \emptyset$, there exists $f: \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$

s.t. $f(A) \in A$ ("choice function")

Proof: (AC) $\Rightarrow$ (AC'). Let $I = \mathcal{P}(X) \setminus \{\emptyset\}$, and

$$(AC) \text{ provides } (x_A)_{A \in I} \in \prod_{A \in I} A$$

let $f(A) = x_A$, done.

(AC') $\Rightarrow$ (AC). Given $\{X_i\}_{i \in I}$, non-empty set

let $X = \bigsqcup_{i \in I} X_i$ (gather all sets together, distinguishing $x_i \neq x_j$ if $i \neq j$)

(AC') provides a choice function $f: \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$.

Then $(f(X_i))_{i \in I} \in \prod_{i \in I} X_i$ $\square$.

Relation, order, equivalence.

A relation on a non-empty set X is subset

$$R \subseteq X \times X = \{(x, y) : x, y \in X\}$$

Write $x R y \Leftrightarrow (x, y) \in R$

We say that a relation R on X is

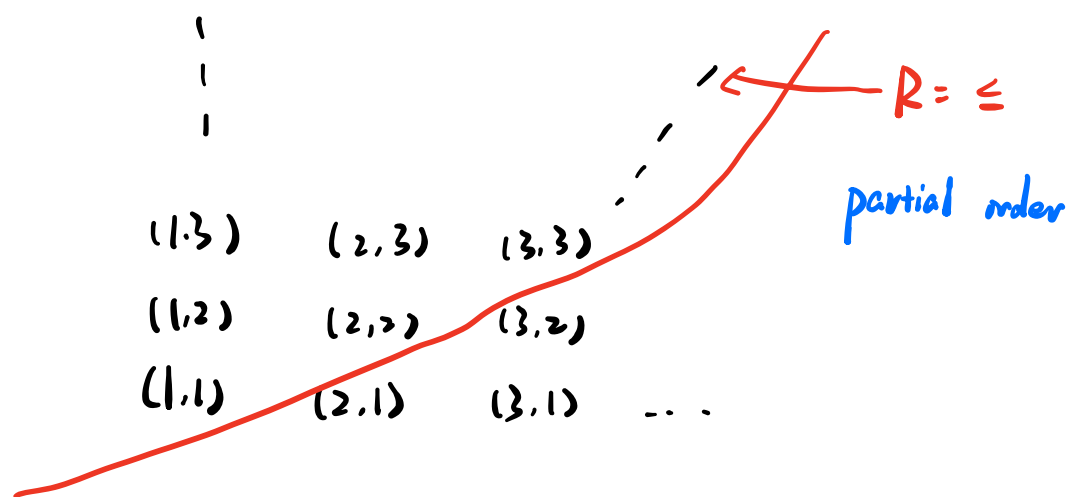
<u>equivalence</u> <u>relation</u>	{	• symmetric : if $x R y \Leftrightarrow y R x$	{	<u>partial</u> <u>order</u> Symbols: $\leq, \leq$.
		• reflexive : if $x R x, \forall x \in X$		
		• transitive : if $x R y$ & $y R z \Rightarrow x R z$.		
Symbols: $\sim, \cong, \dots$		• anti-symmetric : if $x R y$ & $y R x \Rightarrow x = y$.		

Ex: (i) A function on X is a relation: $f \subseteq X \times X$.

so for each x in X , there is one y in X so $(x, y) \in f$.

write $y = f(x)$, $f: X \rightarrow X$

(ii) On $\mathbb{N}$, write $m \leq n \Leftrightarrow$ there is $p \in \{0\} \cup \mathbb{N}$ st. $n = p + m$.



(iii) On $\mathbb{Q} = \left\{ \frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N} \right\}$

$$\frac{m}{n} \leq \frac{p}{q} \Leftrightarrow mq \leq pn \quad \text{partial order.}$$

(iv) On $\mathcal{P}(X)$, $A R B \Leftrightarrow A \subseteq B$

$$R = \subseteq \quad \text{partial order}$$

(v) On $\mathbb{Z} \times \mathbb{N}$, let $(m,n) R (p,q) \Leftrightarrow \underbrace{mp = nq}_{\frac{m}{n} = \frac{p}{q}}$

Def'n: If R is an equivalence relation on X and $x \in X$.

write $[x] = \{y \in X : x R y\}$. and this is called the
equivalence class of x w.r.t. R .

Def'n: We define for an equivalence relation R on X the quotient.

$$X/R = \{[x] : x \in X\} \subseteq \mathcal{P}(X)$$

In (v), $\mathbb{Z} \times \mathbb{N} / \sim$ is represented by $\mathbb{Q}$.

Cantor's Construction of real numbers

Motivation: $\pi = 3.1415 \dots$

$$= \lim_{n \rightarrow \infty} v_n, \quad v_n = \underbrace{3.14 \dots}_n \text{ digits}$$

$$= \sum_{k=0}^{\infty} \frac{4(-1)^k}{k+1} = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{4(-1)^k}{k+1}$$

Moral: We understand real numbers by their rational approximations

Def'n: If $\leq$ is a total order,

then $x, y \in X$ (X with $\leq$), either $x \leq y$ or $y \leq x$

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$'s partial order are total order.

Work on $\mathbb{Q}$

Absolute Value: $r = \frac{m}{n}$, $m \in \mathbb{Z}$, $n \in \mathbb{N}$

$$|r| = \begin{cases} \frac{m}{n}, & m \in \{0\} \cup \mathbb{N} \\ -\frac{m}{n}, & \underbrace{m < 0}_{\substack{m < 0 \Leftrightarrow \\ m \neq 0}} \text{ in } \mathbb{Z} \end{cases}$$

Write $\mathbb{Q}^+ = \{r \in \mathbb{Q}, r > 0\}$

We define Cauchy Sequence of $\mathbb{Q}$:

$$C = \left\{ (f_n) = (f_n)_{n \geq 1}^{\infty} : \begin{array}{l} \text{for any } \varepsilon \in \mathbb{Q}^+, \text{ there is } n_\varepsilon \in \mathbb{N} \\ \text{so } \forall n, m \geq n_\varepsilon \Rightarrow |f_m - f_n| < \varepsilon \end{array} \right\}$$

Note: If $(f_n) \in C$. letting for $\varepsilon \in \mathbb{Q}^+$, n_ε be as above.

we find that $|f_n| \leq \max\{|f_1|, \dots, |f_{n_\varepsilon-1}|, |f_{n_\varepsilon}| + \varepsilon\}$

Equivalence relation: $(f_n), (f'_n) \in C$. write $(f_n) \sim (f'_n)$

if for any $\varepsilon \in \mathbb{Q}^+$, there is n_ε in $\mathbb{N}$ s.t. $|f_n - f'_n| < \varepsilon, \forall n \geq n_\varepsilon$.

"tail equivalence"

check: $\sim$ is an equivalence relation on C .

Proposition (Arithmetic) let $(f_n) \sim (f'_n), (r_n) \sim (r'_n)$ in C .

(i) $(f_n + r_n) \in C$, and $(f_n + r_n) \sim (f'_n + r'_n)$

(ii) $(f_n r_n) \in C$ and $(f_n r_n) \sim (f'_n r'_n)$

(iii) if $(f_n) \neq (0) = (0, 0, 0, \dots)$, then there is $(s_n) \in C$

s.t. $(f_n s_n) = (1) = (1, 1, 1, \dots)$

proof: (ii) Assign $\#1$, (i) is similar to but easier than (ii)

(iii) is similar to (ii).

Note:
$$S_n = \begin{cases} \frac{1}{f_n}, & \text{if } f_n \neq 0 \\ 1, & \text{if } f_n = 0 \leftarrow \text{finitely many.} \end{cases}$$

□.

We define the real numbers by

$$\mathbb{R} = \mathbb{C} / \sim, \quad \bar{(f_n)} = \bar{f_n} - \text{equivalence class.}$$

$$\text{i.e. } \mathbb{R} = \{ \bar{f_n} : (f_n) \in \mathbb{C} \}$$

The proposition tells me that operations

$$\bar{f_n + r_n} = \bar{f_n} + \bar{r_n}, \quad \bar{f_n \cdot r_n} = \bar{f_n} \cdot \bar{r_n} \text{ are well-defined.}$$

i.e. makes sense on equivalence classes.

Furthermore, $\bar{f_n} \neq \bar{0} = \bar{(0, 0, 0, \dots)}$, then $\exists \bar{s_n} \in \mathbb{R}$ s.t.

$$\bar{f_n} \bar{s_n} = \bar{f_n s_n} = \bar{1} = \bar{(1, 1, \dots)}$$

i.e. we get all expected arithmetic operations.

Notation: Addictive inverse: $-\bar{f}_n = \bar{-f}_n$

Multiplicative inverse: $\bar{f}_n \neq 0, \bar{f}_n^{-1} = \bar{s}_n, \bar{s}_n$ as above.

Order in $\mathbb{R}$: $\bar{f}_n, \bar{s}_n \in \mathbb{R}$.

Write $\bar{f}_n \leq \bar{s}_n$ if there is $n_0 \in \mathbb{N}$ s.t. $f_n \leq s_n, \forall n \geq n_0$

or $\bar{f}_n = \bar{s}_n$

Hence write $\bar{f}_n < \bar{s}_n$ if $\bar{f}_n \neq \bar{s}_n$, and there is $n_0 \in \mathbb{N}$

s.t. $f_n < s_n, \forall n \geq n_0$.

Goal: $\leq$ is a total order.

Proposition: Let $\mathbb{R}^+ = \{\bar{f}_n \in \mathbb{R}, \bar{0} < \bar{f}_n\}$ ("positive elements")

Then (i) $\bar{f}_n, \bar{s}_n \in \mathbb{R}^+ \Rightarrow \bar{f}_n + \bar{s}_n, \bar{f}_n \bar{s}_n \in \mathbb{R}^+$

(ii) $\mathbb{R} = (-\mathbb{R}^+) \cup \{\bar{0}\} \cup (\mathbb{R}^+)$, disjoint union.

(iii) $\bar{f}_n < \bar{s}_n$ in $\mathbb{R} \Leftrightarrow \bar{s}_n - \bar{f}_n \in \mathbb{R}^+$

Proof: (i): Straight Forward. (we borrow from order structure of $\mathbb{Q}$)

(ii): Assign #1.

(iii): $[f_n] < [r_n] \Leftrightarrow [f_n] \neq [r_n], f_n \in r_n, \forall n \geq n_0$

$\Leftrightarrow [f_n - r_n] \neq 0$ and $0 \leq r_n - f_n, \forall n \geq 0$

$\Leftrightarrow [r_n - f_n] \in \mathbb{R}^+$

□.

Corollary: $\leq$ is a total order on $\mathbb{R}$.

proof: let $[f_n], [r_n] \in \mathbb{R}$ and $[f_n] \neq [r_n]$

Then by (ii) above, we have

either $[r_n - f_n] \in \mathbb{R}^+ \Leftrightarrow [f_n] < [r_n]$ (by (iii)) $\Rightarrow [f_n] \leq [r_n]$

or $[f_n - r_n] \in \mathbb{R}^+ \Leftrightarrow [r_n] < [f_n]$ (by (iii)) $\Rightarrow [r_n] \leq [f_n]$

□.

Binary representation: let $x \in \mathbb{R}$. Really, $x = [f_n]$.

Recall that (f_n) is bounded, i.e., $f_n \leq |f_n| \leq M$, for some $M \in \mathbb{Q}$.

and hence $x \leq M = [M, M, \dots]$

Thus, there is $N \in \mathbb{N}$ s.t. $x < N$ [$M = \frac{m}{n}$, $m, n \in \mathbb{N}$, $M > 0 \Rightarrow M \leq n$]

Integer floor: $\lfloor x \rfloor = \min \{ n \in \mathbb{N} : x < n \} - 1$

property of $\mathbb{N}$ guarantee this to exist.

$\mathbb{N}$ is well-ordered.

So $\lfloor x \rfloor = \{0\} \cup \mathbb{N}$, $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$

Hence $0 \leq x - \lfloor x \rfloor < 1$.

We define $x_1 = \begin{cases} 0, & \text{if } x - \lfloor x \rfloor < \frac{1}{2} \\ 1, & \text{if } x - \lfloor x \rfloor \geq \frac{1}{2} \end{cases}$

$x_2 = \begin{cases} 0, & \text{if } x - (\lfloor x \rfloor + \frac{x_1}{2}) < \frac{1}{2} \\ 1, & \text{if } x - (\lfloor x \rfloor + \frac{x_1}{2}) \geq \frac{1}{2} \end{cases}$

$\vdots$

$x_{n+1} = \begin{cases} 0, & \text{if } x - (\lfloor x \rfloor + \frac{x_n}{2}) < \frac{1}{2} \\ 1, & \text{if } x - (\lfloor x \rfloor + \frac{x_n}{2}) \geq \frac{1}{2} \end{cases}$

Fact (Exercise):

$(\lfloor x \rfloor + \frac{x_1}{2}, \lfloor x \rfloor + \frac{x_1}{2} + \frac{x_2}{2}, \dots) \in \mathcal{C}$ (Cauchy sequence of $\mathbb{Q}$)

$$\text{and } x = \left[\lfloor x \rfloor + \frac{x_1}{2}, \lfloor x \rfloor + \frac{x_1}{2} + \frac{x_2}{2}, \dots \right]$$

$$\text{Notation: } x = \lfloor x \rfloor + \sum_{k=1}^{\infty} \frac{x_k}{2^k} = \lim_{n \rightarrow \infty} \left(\lfloor x \rfloor + \sum_{k=1}^n \frac{x_k}{2^k} \right)$$

$$= \lfloor x \rfloor . x_1 x_2 x_3 \dots$$

Bounds: let $(X, \leq)$ be a partially ordered (p.o.) set.

i.e. X is a non-empty set, $\leq$ is a partial order on X .

let $A \subseteq X$, $A \neq \emptyset$. We say that

- A admits an upper-bound (u.b.) if there is u in X s.t. $a \leq u$ for $a \in A$.
- A admits a supremum if there exists an u.b. s for A s.t. for any u.b. u of A , $s \leq u$.

We say that $(X, \leq)$ is complete if every non-empty subset, admitting an u.b., admits a supremum.

Ex: i), $(\mathcal{P}(X), \subseteq)$ ($X \neq \emptyset$).

$$\emptyset \neq A = \{A_i\}_{i \in I} \subseteq \mathcal{P}(X)$$

Here $\bigcup_{i \in I} A_i = A$ is the supremum of A .

i.e. $(\mathcal{P}(X), \subseteq)$ is complete.

(ii) Consider $(\mathbb{Q}, \leq)$ (usual ordering)

$\mathbb{N} \subseteq \mathbb{Q}$ has no u.b.

$A = \{q \in \mathbb{Q} : 0 < q, q^2 < 2\}$ admits u.b. (say $u=2$), but no supremum.

Note if, for $A \in \mathcal{X}$, $(X, \leq)$ poset) a supremum exists,

it is unique. (use anti-symmetric)

Write $\sup A$ as supremum of A .

Ex: $(\mathbb{R}, \leq)$ (usual order) is complete.

sketch: let $\emptyset \neq A \subseteq \mathbb{R}$ admits an u.b.

Pick z_0, r_0 in $\mathbb{Q}$ s.t.

- z_0 is not an u.b. of A .
- r_0 is an u.b. of A .

Inductively, let

$$r_n = \begin{cases} \frac{1}{2}(z_{n-1} + r_{n-1}), & \text{if } \frac{1}{2}(z_{n-1} + r_{n-1}) \text{ is an u.b.} \\ r_{n-1}, & \text{o.w.} \end{cases}$$

$$z_n = \begin{cases} z_{n-1}, & \text{if } \frac{1}{2}(z_{n-1} + r_{n-1}) \text{ is an u.b.} \\ \frac{1}{2}(z_{n-1} + r_{n-1}), & \text{if } \frac{1}{2}(z_{n-1} + r_{n-1}) \text{ is not an u.b.} \end{cases}$$

Then $(z_n) \sim (r_n)$ in C , and $s = \bar{z}_n = \bar{r}_n$

is THE supremum of A .

Remark Analogously, we can define lower bounds and infimum for a set in poset

Prop: If $(X, \leq)$ is a complete poset, and $\emptyset \neq A \subseteq X$ admits a lower bound, then A admits an infimum.

Proof: let $L = \{l \in X, l \leq a, \forall a \in A\}$

Then $L \neq \emptyset$ (by assumption on A) and L admits an u.b.

let $s = \sup L$. Then $s \leq a, \forall a \in A$.

Indeed, if there were any a_0 in A st.

$a_0 < s$ ($a_0 \leq s, a_0 \neq s$), then $l \leq a_0$ for all l in L .

This violates the def'n of $s = \sup L$.

Hence $s = \inf(A)$

□.

Chains and Zorn's Lemma:

let $(X, \leq)$ be a poset.

• A chain in $(X, \leq)$ is any non-empty subset on which $\leq$ is a total order.

• A element m of X is maximal if $m \leq x, \forall x \in X$
 $\Rightarrow \text{max.}$

Zorn's Lemma (Axiom, equivalent to AC)

If in a poset $(X, \leq)$, each chain admits an u.b.

then X admits a maximal element.

Recall: If V is a vector space over a field K , then a subset

• $L \subseteq V$ is lin. indep. if $\forall$ finite $\{v_1, \dots, v_n\} \subseteq L$ (disjoint pts)

the equation $\sum_{k=1}^n \alpha_k v_k = 0$, where $\alpha_k \in K$, has unique sol'n $\alpha_1 = \dots = \alpha_n = 0$

• $S \subseteq V$ is spanning if $\forall v \in V$, we can write $v = \sum_{k=1}^n \alpha_k v_k$, $v_k \in S$

• $B \subseteq V$ is a basis if it both lin. indep. and spanning.

Theorem: let V be a vector space over K . $L \subseteq V$ be a non-empty lin. indep. subset. Then $\exists$ basis B s.t. $L \subseteq B$.

proof (Zorn's lemma): let $\mathcal{L} = \{L' \subseteq V : L \subseteq L', L' \text{ is lin. indep.}\}$
 $\subseteq \mathcal{P}(V)$

Notice that $L \in \mathcal{L}$, so $\mathcal{L} \neq \emptyset$.

let's consider p.o. of inclusion, $\subseteq$, on $\mathcal{L}$.

Suppose $\mathcal{C} \subseteq \mathcal{L}$ is a chain (i.e. non-empty total order subset)

Let $U = \bigcup_{C \in \mathcal{C}} C \in \mathcal{P}(V)$.

Then $L \subseteq U$ (obvious, $L \subseteq C$ for each $C \in \mathcal{C}$)

$\cdot U$ is lin. indep.

To see linear independence, consider $\{v_1, \dots, v_n\} \subseteq U$ (distinct p-1)

Each $v_k \in U = \bigcup_{C \in \mathcal{C}} C$, so there is $C_k \in \mathcal{C}$, so $v_k \in C_k$.

Since $\mathcal{C}$ is a chain, up to relabelling, we may assume

$$C_1 \subseteq C_2 \subseteq \dots \subseteq C_n.$$

Hence $\{v_1, \dots, v_n\} \subseteq C_n$, and $C_n \in \mathcal{L}$, i.e., lin. indep.

$$\text{Hence } 0 = \sum_{k=1}^n \alpha_k v_k, \alpha_k \in \mathbb{F} \Rightarrow \alpha_1 = \dots = \alpha_n = 0$$

So we have proved that U is lin. indep.

i.e. $U \in \mathcal{L}$, so $\mathcal{C}$ admits U as an upper bound.

Thus, Zorn's lemma tells that $\mathcal{L}$ admits a maximal element, B .

If B spans V , we are done. Otherwise, there exists

v_0 in V s.t. v_0 is not a lin. comb. of elements of B .

Let $B' = B \cup \{v_0\}$. Consider the eq'n: $\alpha_0 v_0 + \sum_{k=1}^n \alpha_k v_k = 0$

where $\{v_1, \dots, v_n\} \subseteq B$. (distinct).

Case #1: $\alpha_0 \neq 0 \Rightarrow v_0 = \sum_{k=1}^n \left(-\frac{\alpha_k}{\alpha_0}\right) v_k$, ~~to~~ $v_0 \notin \text{Span } B$.

Case #2: $\alpha_0 = 0 \Rightarrow \alpha_0 v_0 + \sum_{k=1}^n \alpha_k v_k = \sum_{k=1}^n \alpha_k v_k = 0 \Rightarrow \alpha_1 = \dots = \alpha_n = 0$

$\Rightarrow B'$ is lin. indep. Since $L \subseteq B \subseteq B'$, $B' \in L$

~~#~~ to B is maximal element

Thus, B spans V , and hence B is a basis

$\square$.

Cardinal numbers

Let X, Y be non-empty sets. A function $f: X \rightarrow Y$ ($f = Y^X$)

is injective if it $x_1 \neq x_2 \in X, \Rightarrow f(x_1) \neq f(x_2)$

• surjective if $\forall y \in Y, \exists x \in X$ s.t. $f(x) = y$.

• bijective if $f: X \rightarrow Y$ is both injective and surjective.

If $f: X \rightarrow Y$ is bijection, there exists $g: Y \rightarrow X$

so $f \circ g = \text{id}_Y$, $g \circ f = \text{id}_X$.

This is unique, say $g = f^{-1}$

Cardinal Equation of Sets: $X \approx Y$, if there is a bijection $f: X \rightarrow Y$

Write $X \underset{f}{\approx} Y$.

Notice that $\cdot X \underset{f}{\approx} Y \Rightarrow Y \underset{f^{-1}}{\approx} X$

• $X \underset{\text{id}_X}{\approx} X$

• $X \underset{f}{\approx} Y, Y \underset{g}{\approx} Z \Rightarrow X \underset{g \circ f}{\approx} Z$.

This is an equivalence relation of set. Given a set A ,

write $|A|$ for its equivalence class, called a cardinal number

• Standard ones: $|\mathbb{N}| = \aleph_0$ ("aleph naught")

$$|\mathbb{R}| = \mathfrak{c} \text{ ("continuum")}$$

$$\left. \begin{array}{l} |\{1, \dots, n\}| = n \\ |\emptyset| = 0 \end{array} \right\} \text{finite}$$

Ex: (i) $\mathbb{N} \approx \mathbb{Z}$, $f(n) = \begin{cases} \frac{1}{2}(1-n), & n \text{ odd} \\ \frac{1}{2}n, & n \text{ even} \end{cases}$

$$\Rightarrow |\mathbb{Z}| = |\mathbb{N}| = \aleph_0.$$

(ii) $\mathbb{R} \underset{f}{\approx} (-1, 1)$, $f(x) = \frac{x}{|x|+1}$ (compute inverse)

(iii) $(0, 1) \underset{h}{\approx} (a, b)$, $a < b \in \mathbb{R}$, $h(t) = a + t(b-a)$

Conclusion: $\mathbb{R} \approx (-1, 1) \approx (0, 1) \approx (a, b)$

$$\Rightarrow |(a, b)| = \mathfrak{c}, \quad a < b \in \mathbb{R}.$$

Arithmetic : let A, B be sets.

Addition: $|A| + |B| = |A \sqcup B|$

Multiplication: $|A| |B| = |A \times B|$

Exponent : $|A|^{|B|} = |A^B|$, if $B \neq \emptyset$.

where $A^B = \{f: B \rightarrow A\}$ functions from B to A .

$$= \{(a_b)_{b \in B} : \text{each } a_b \in A\} \quad (f(b) = a_b)$$

Properties (Commutativity)

$$|A| + |B| = |B| + |A| \text{ , i.e., } A \sqcup B \approx B \sqcup A$$

$$|A| |B| = |B| |A| \text{ , i.e., } A \times B \approx B \times A$$

(Distributive Law)

$$|A| (|B| + |C|) = |A| |B| + |A| |C| \text{ , i.e., } A \times (B \sqcup C) = (A \times B) \sqcup (A \times C)$$

(Associativity)

$$(|A| + |B|) + |C| = |A| + (|B| + |C|)$$

$$(|A| |B|) |C| = |A| (|B| |C|)$$

(Exponentiation)

$$|A|^{(|B|+|C|)} = |A|^{(|B|)} |A|^{(|C|)}, \quad A^{B \cup C} \approx A^B \times A^C$$

$$\text{via } f \mapsto (f|_B, f|_C)$$

$$|A|^{(|B||C|)} = (|A|^{(|B|)})^{(|C|)}, \quad A^{B \times C} \approx (A^B)^C$$

$$\text{via } f \mapsto (f(b, \cdot))_{b \in B}$$

$\uparrow$
 $c \text{ in } C \text{ goes here}$

Sample computations:

$$1 + \mathcal{X}_0 = |\{x_0\} \sqcup \emptyset| = \mathcal{X}_0$$

$\uparrow$ invented a point

$$\{x_0\} \sqcup \mathbb{N} \approx \mathbb{N}, \quad f: \{x_0\} \sqcup \mathbb{N} \rightarrow \mathbb{N}$$

$$f(x) = \begin{cases} 1 & , x = x_0 \\ x+1 & , x \in \mathbb{N} \end{cases}$$

"Hilbert hotel".

Induction $\Rightarrow n + \aleph_0 = \aleph_0$, n finite

$$\aleph_0 + \aleph_0 = |\mathbb{N} \sqcup \mathbb{N}|$$

$$\mathbb{N} \sqcup \mathbb{N} = \{1, 2, 3, \dots\} \cup \{1_2, 2_2, 3_2, \dots\}$$

$$f: \mathbb{N} \sqcup \mathbb{N} \rightarrow \mathbb{N}, \quad f(n_k) = \begin{cases} 2n_k, & k=1, \quad k=1 \\ 2(n_k-1)+1, & k=2 \end{cases}$$

Comparison: let A, B be sets

$A \leq B$, if there is an injective $f: A \rightarrow B$

We may write $A \leq_f B$

Note that $A \leq_{\text{id}} A$; $A \leq_f B, B \leq_g C \Rightarrow A \leq_{gf} C$

Question: If $A \leq B$, $B \leq A$, is $A \approx B$?

Cantor - Bernstein - Schroder Theorem:

If A, B are sets,

$$A \leq B \text{ and } B \leq A$$

then $A \approx B$

Ex: (i) let $a < b$ in $\mathbb{R}$.

Then $[a, b] \approx \mathbb{R}$.

proof: $[a, b] \leq \mathbb{R}$ (obvious)

Conversely, $\mathbb{R} \approx (a, b) \leq [a, b]$

By CBS Theorem, $[a, b] \approx \mathbb{R}$, i.e. $|[a, b]| = c$.

□.

Alternatively, $(a, b) \geq \underbrace{\left\{ a + \frac{b-a}{n}, n=2, 3, \dots \right\}}_C$

$$|C|: \aleph_0 = \aleph_0 + 1.$$

Alternatively, (Faces of continuum)

$$[0,1] \underset{\beta}{\leq} \{0,1\}^{\mathbb{N}}$$

$$x \in [0,1], \quad x = \sum_{k=1}^{\infty} \frac{x_k}{2^k} \text{ (binary)}, \text{ never } x_k = 1, \forall k \geq n.$$

$$\beta(x) = (x_k)_{k=1}^{\infty} \in \{0,1\}^{\mathbb{N}}$$

$$\text{Conversely, } \{0,1\}^{\mathbb{N}} \underset{\tau}{\leq} [0,1], \quad \tau((x_k)_{k=1}^{\infty}) = \sum_{k=1}^{\infty} \frac{x_k}{2^k} \text{ (binary)}$$

$$\text{CBS Thm} \Rightarrow \{0,1\}^{\mathbb{N}} \approx [0,1]$$

$$\text{Thus, } 2^{\aleph_0} = |\{0,1\}^{\mathbb{N}}| = |[0,1]| = \mathfrak{c}.$$

$$\mathcal{P}(\mathbb{N}) \underset{f}{\approx} \{0,1\}^{\mathbb{N}}$$

$$\text{let } A \in \mathcal{P}(\mathbb{N}); \quad f(A) = x_A, \quad x_A(n) = \begin{cases} 1, & n \in A \\ 0, & n \notin A \end{cases}$$

Exercise: This is bijective.

Hence $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}| = \mathfrak{c}$.

Composition Theorem

If A, B are sets, then at least one of the following holds:

$$A \leq B, \text{ or } B \leq A.$$

Consequence: at least one of

$$\underbrace{|A| \leq |B|}_{A \leq B} \text{ or } |B| \leq |A| \text{ is true.}$$

Prop'n (Surjectivity)

If X, Y are sets which admits a surjection $\varphi: X \rightarrow Y$

$$\text{Then } |Y| \leq |X|$$

Proof: let $f: \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$ be a choice function.

$$\text{i.e. } f(A) \in A. \quad (A \subset X)$$

$$\text{let } g: Y \rightarrow X ; g(y) = f(\varphi^{-1}(\{y\}))$$

check that f is injective

$$\text{i.e. } Y \underset{f}{\leq} X, \text{ i.e. } |Y| \leq |X| \quad \square.$$

Comparison Theorem: If X, Y are non-empty sets,

$$\text{then either } |X| \leq |Y|, |Y| \leq |X|$$

Sketch of proof: let $\Delta = \{(A, f) : A \subseteq X, f: A \rightarrow Y \text{ is injective}\}$

Note that $\Delta \neq \emptyset$ since A can be singleton

Partial Order on Δ : $(A, f) \leq (B, g)$
if $A \subseteq B, g|_A = f.$ } check!

let $\Gamma \subseteq \Delta$ be a chain.

Exercise: $U = \bigcup_{(C, f) \in \Gamma} C \subseteq X. g: U \rightarrow Y, g(x) = f(x) \text{ if } x \in C.$
 $(C, f) \in \Gamma$

has that $(V, g) \in \Delta$ and is an u.b. for Γ .

Zorn's lemma tells us maximal element (M, f) for Δ exists.

Two potential cases:

- $M = X$, we're done.
- $f(M) = Y$, we're done. (Use Surjective prop'n
or inverse function
 $f^{-1}: Y \rightarrow M \subseteq X$)

Alternatively, neither of the two holds.

$$\exists x_0 \in X \setminus M, y_0 \in Y \setminus f(M)$$

$$\text{Define } g: M \cup \{x_0\} \rightarrow Y, \quad g(x) = \begin{cases} f(x), & x \in M \\ y_0, & x = x_0. \end{cases}$$

$$\Rightarrow (M, f) \leq (M \cup \{x_0\}, g)$$

but $(M, f) \neq (M \cup \{x_0\}, g)$, which ~~is~~ maximality. $\square$.

Theorem (Cantor) : Given a set X , $|X| < |\mathcal{P}(X)|$

i.e. $X \leq \mathcal{P}(X)$ but $\mathcal{P}(X) \neq X$

Proof: First, if $X \neq \emptyset$, then

$$|X|=0 \quad , \quad |\mathcal{P}(\emptyset)| = |\{\emptyset\}| = 1.$$

Now suppose that $X \neq \emptyset$. Then

$x \mapsto \{x\} : X \rightarrow \mathcal{P}(X)$ is injective so $X \leq \mathcal{P}(X)$

Now suppose that $f: X \rightarrow \mathcal{P}(X)$ is any function.

Let $E = \{x \in X : x \notin f(x)\} \in \mathcal{P}(X)$.

If we had $E \in f(X)$ (e.g. f surjective)

i.e. $E = f(x)$ for some $x \in X$.

Consider . if $x \in E$, then $x \notin f(x) = E$, by choice of x #.

• if $x \notin E$, then $x \in f(x) = E$, by def'n of E #.

This contradiction shows that $E \neq f(x)$

Thus, no surjection $f: X \rightarrow P(X)$ exists.

So no bijection exists.

□.

Corollary: $\aleph_0 < c$.

proof: $\aleph_0 = |N| < |P(N)| = c$

□.

Remark: Russel's Paradox:

If there were a set of all sets U

which were a set $|U| < |P(U)|$. Problem.

Moral: We don't discuss the "set of all sets".

Prop'n: (Infinite sets)

let A be a set, Then $\neg \exists A \in$.

$$(i): n \leq |A|, \forall n \in \mathbb{N}$$

$$(ii): \aleph_0 \leq |A|$$

$$(iii): 1 + |A| = |A| \text{ (Hilbert Hotel)}$$

(i) $\Rightarrow$ (ii): let $\varphi_n: \{1, \dots, n\} \rightarrow A$ be an injective function

for each $n \in \mathbb{N}$. Define $f: \mathbb{N} \rightarrow A$ by

$$f(n) = \varphi_n(1)$$

$$n \geq 2, \quad f(n) = \varphi_{n+1}(k), \text{ where } k = \min \{j \in \{1, \dots, n\} \text{ s.t. } \varphi_{n+1}(j) \in \{f(n), \dots, f(1)\}\}.$$

(ii) $\Rightarrow$ (iii): Consider $x_0 \notin A$.

There is an injection of $f: \mathbb{N} \rightarrow A$.

Define $g: \{x_0\} \cup A \rightarrow A$

$$\text{by } g(x) = \begin{cases} f(n) & , x = x_0 \\ f(n+1) & , x = f(n) \end{cases} \text{ is bijection.}$$

$x \quad \text{a.w.}$

$$(iii) \Rightarrow (i): \{x_0\} \cup A \approx A.$$

Then $f(A) \subsetneq A$, $f \circ f(A) \subsetneq f(A)$, etc.

Then $|A \setminus f^{\circ n}(A)| \geq n$.

where $f^{\circ n} = \underbrace{f \circ \dots \circ f}_{n \text{ times}}$ $\square$.

Metric Spaces :

Note Having been constructed, let's take $\mathbb{R}$
for granted $\cdot \leq, <, +, \cdot, |\cdot|$, etc.

IMPORTANT: completeness. (See Supplement Notes)

Def'n : Let X be a non-empty set.

A metric is a function

$$d: X \times X \longrightarrow \mathbb{R}$$

Sit. for x, y, z in X , we have

$$\text{(non-negativity)} \quad d(x, y) \geq 0$$

$$\text{(non-degeneracy)} \quad d(x, y) = 0 \Leftrightarrow x = y.$$

$$\text{(symmetric)} \quad d(x, y) = d(y, x)$$

$$\text{(triangle inequality)} \quad d(x, z) \leq d(x, y) + d(y, z)$$

Conclusion: d defines a "distance" on X .

We call (X, d) a metric space

Ex: (i) on $\mathbb{R}$, $d(x, y) = |x - y|$

(ii) (Trivial metric) let X be a non-empty set.

Define $d: X \times X \rightarrow \{0, 1\} \subseteq \mathbb{R}$

$$d(x, y) = \begin{cases} 1, & \text{if } x \neq y \\ 0, & \text{if } x = y \end{cases} \quad (\text{check } \triangle \text{ ineq})$$

(iii) let $f: \mathbb{R} \rightarrow \mathbb{R}$ which is strictly increasing

$x < y$ in $\mathbb{R} \Rightarrow f(x) < f(y)$, define

$$d_f: \mathbb{R} \rightarrow \mathbb{R}, d_f(x, y) = |f(x) - f(y)|$$

(iv) (SNCF metric) let X be a non-empty set,

$$f: X \rightarrow [0, \infty) \text{ s.t.}$$

$$f^{-1}(\{0\}) = \{p_0\} \text{ (only } p_0 \text{ satisfies } f(p_0) = 0)$$

$$\text{let } d_f: X \times X \rightarrow \mathbb{R}, d_f(x, y) = \begin{cases} f(x) + f(y), & x \neq y \\ 0, & x = y \end{cases}$$

$$\Delta \text{ inequality: } d(x, z) = f(x) + f(z)$$

$$\leq f(x) + f(y) + f(y) + f(z)$$

$$= d(x, y) + d(y, z)$$

(only equality is achieved iff $y = p_0$)

(v). X be a non-empty set,

$$\mathcal{F}(X) = \{F \in \mathcal{P}(X), F \text{ is finite}\}$$

let

$$P: \mathcal{F}(X) \times \mathcal{F}(X) \rightarrow \mathbb{R}.$$

$$P(E, F) = |E \Delta F|, \text{ where } E \Delta F = (E \setminus F) \cup (F \setminus E)$$

Def'n: Let V be a vector space over $\mathbb{R}$.

A norm is a function $\|\cdot\|: V \rightarrow \mathbb{R}$ s.t.

for $x, y \in V, \alpha \in \mathbb{R}$

(non-negativity) : $\|x\| \geq 0$

(non-degeneracy) : $\|x\| = 0 \Leftrightarrow x = 0$

($|\cdot|$ -homogeneity) : $\|\alpha x\| = |\alpha| \cdot \|x\|$

(sub-additivity) : $\|x+y\| \leq \|x\| + \|y\|$

Ex: $\mathbb{R}$ is a vector space over it self.

$|\cdot|$ is a norm on $\mathbb{R}$.

Metric from norm: If $(V, \|\cdot\|)$ is a normed (vector) space,

$$d(x, y) = \|x - y\|.$$

Some norms on $\mathbb{R}^n$: $x \in \mathbb{R}^n$, $x = (x_1, \dots, x_n)$, each $x_j \in \mathbb{R}$.

$$\text{write } \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$$

$$\|x\|_1 = |x_1| + \dots + |x_n|$$

p-norms: let $p \in (1, \infty)$, and define for $x \in \mathbb{R}^n$,

$$\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{\frac{1}{p}}$$

Goal: Show that $\|\cdot\|_p$ defines a norm on $\mathbb{R}^n$.

Lemma (Young's Inequality)

Let $\alpha, \beta \geq 0$ in $\mathbb{R}$, $p \in (1, \infty)$, and f is

sit. $\frac{1}{p} + \frac{1}{q} = 1$ (Hölder conjugate)

So $q = \frac{p}{p-1} > 1$. Then

$$\alpha\beta \leq \frac{\alpha^p}{p} + \frac{\beta^q}{q}$$

proof : if $\alpha\beta = 0$, then it's true.

Hence assume $\alpha, \beta \neq 0 \Rightarrow \alpha, \beta > 0$.

let $g: (0, +\infty) \rightarrow \mathbb{R}; x \mapsto \frac{x^p}{p} + \frac{1}{q} - x$

$$\text{then } g'(x) = x^{p-1} - 1 = \begin{cases} < 0, & \text{if } x < 1 \\ > 0, & \text{if } x > 1 \end{cases}$$

$\Rightarrow g$ decreases on $(0, 1)$ and increasing on $(1, +\infty)$

$\Rightarrow \exists!$ minimum for g at $x=1$

with $g(1) = \frac{1}{p} + \frac{1}{q} - 1 = 0$. i.e., $g(x) \leq 0 \Leftrightarrow x=1$

let $x = \frac{\alpha}{\beta^{2/p}}$, then

$$0 \leq g\left(\frac{\alpha}{\beta^{2/p}}\right) = \frac{\alpha^p}{p \cdot \beta^2} + \frac{1}{2} - \frac{\alpha}{\beta^{2/p}}$$

$$\begin{aligned} \Rightarrow 0 &\leq \frac{\alpha^p}{\beta^2} + \frac{\beta^2}{2} - \alpha \cdot \beta^{2-\frac{2}{p}} \\ &= \frac{\alpha^p}{\beta^2} + \frac{\beta^2}{2} - \alpha \cdot \beta \end{aligned}$$

Hölder's Inequality. let $x, y \in \mathbb{R}^n$, $p \in (1, +\infty)$

let q be the conjugate index $\frac{1}{p} + \frac{1}{q} = 1$.

Then

$$\left| \sum_{j=1}^n x_j y_j \right| \leq \sum_{j=1}^n |x_j| |y_j| \leq \|x\|_p \cdot \|y\|_q.$$

↑
D. def. of l_1 .

Proof: let consider the second $\leq$ only.

This is trivial if $x=0$ or $y=0$.

Consider $x, y \neq 0$, so $\|x\|_p \cdot \|y\|_q \neq 0$.

$$\begin{aligned} \text{Consider } \sum_{j=1}^n \underbrace{\frac{x_j}{\|x\|_p}}_{\alpha_j} \cdot \underbrace{\frac{y_j}{\|y\|_q}}_{\beta_j} &= \sum_{j=1}^n \alpha_j \cdot \beta_j \stackrel{\text{Young}}{\leq} \sum_{j=1}^n \left(\frac{\alpha_j^p}{p} + \frac{\beta_j^q}{q} \right) \\ &= \sum_{j=1}^n \left(\frac{|x_j|^p}{p \cdot \|x\|_p^p} + \frac{|y_j|^q}{q \cdot \|y\|_q^q} \right) \\ &= \frac{1}{p \cdot \|x\|_p^p} \sum_{j=1}^n |x_j|^p + \frac{1}{q \cdot \|y\|_q^q} \sum_{j=1}^n |y_j|^q \\ &= \frac{1}{p} + \frac{1}{q} = 1 \end{aligned}$$

□.

Minkowski's Inequality: If $x, y \in \mathbb{R}^n$, $p \in (1, +\infty)$, then

$$\|x+y\|_p \leq \|x\|_p + \|y\|_p$$

Proof: Let q be the conjugate index to p : $\frac{1}{p} + \frac{1}{q} = 1$

If $x+y=0$, this is trivial. Assume $x+y \neq 0$, so $\|x+y\| \neq 0$

$$\text{Then } \|x+y\|_p^p = \sum_{j=1}^n |x_j+y_j|^p = \sum_{j=1}^n |x_j+y_j| \cdot |x_j+y_j|^{p-1}$$

$$\begin{aligned}
& \leq \sum_{j=1}^n |x_j| \cdot |x_j + y_j|^{p-1} + \sum_{j=1}^n |y_j| \cdot |x_j + y_j|^{p-1} \\
& \stackrel{\text{Hölder}}{\leq} \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^{(p-1)q} \right)^{\frac{1}{q}} + \left(\sum_{j=1}^n |y_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^{(p-1)q} \right)^{\frac{1}{q}} \\
& \leq (\|x\|_p + \|y\|_p) \cdot \|x+y\|_p^{\frac{p}{q}} \\
& \Rightarrow \|x+y\|_p^{p-\frac{p}{q}} \leq \|x\|_p + \|y\|_p \\
& \Rightarrow \|x+y\|_p \leq \|x\|_p + \|y\|_p \quad \square.
\end{aligned}$$

Corollary: $\|\cdot\|_p$ is a norm on $\mathbb{R}^n$.

Proof: Non-negativity, non-degeneracy are obvious.

$$\|\alpha x\| = |\alpha| \cdot \|x\| \quad \checkmark$$

triangle inequality: $\checkmark$

$\square$.

The Space ℓ_p

Note that $\mathbb{R}^{\mathbb{N}}$ (seq space) is $\mathbb{R}$ -vector space

$$(x_k)_{k=1}^{\infty} + (y_k)_{k=1}^{\infty} = (x_k + y_k)_{k=1}^{\infty}$$

$$\alpha \cdot (x_k)_{k=1}^{\infty} = (\alpha \cdot x_k)_{k=1}^{\infty}$$

let for $p \in [1, \infty)$, $\ell_p = \{(x_k)_{k=1}^{\infty} \in \mathbb{R}^{\mathbb{N}} : \|x\|_p < \infty\}$.
 $\uparrow$
 1 is included.

$$\text{where } \|x\|_p = \begin{cases} \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{\frac{1}{p}}, & \text{if series converges} \\ \infty & \text{if series diverges.} \end{cases}$$

Prop'n: $(\ell_p, \|\cdot\|_p)$ is a linear subspace of $\mathbb{R}^{\mathbb{N}}$, and is a normed space.

proof: Again, non-negativity and non-degeneracy are obvious.

let's consider sub-meq. let $x, y \in \ell_p$

$$\|x+y\|_p = \underbrace{\left(\sum_{k=1}^{\infty} |x_k + y_k|^p \right)^{\frac{1}{p}}}_{\text{not } \infty} = \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n |x_k + y_k|^p \right)^{\frac{1}{p}}$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n |x_k + y_k|^p \right)^{\frac{1}{p}}, \text{ if diverges, get } \infty \text{ on both side.}$$

and diverges, use $t \rightarrow t^{\frac{1}{p}}$ is cts on $[0, \infty)$

$$\leq \lim_{n \rightarrow \infty} \left[\left(\sum_{k=1}^n |x_k|^p \right)^{\frac{1}{p}} + \left(\sum_{k=1}^n |y_k|^p \right)^{\frac{1}{p}} \right]$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n |x_k|^p \right)^{\frac{1}{p}} + \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n |y_k|^p \right)^{\frac{1}{p}}$$

$$= \|x\|_p + \|y\|_p$$

Thus, as $x, y \in \mathcal{L}_p$, we see that $x+y \in \mathcal{L}_p$.

$$\text{and } \|x+y\|_p \leq \|x\|_p + \|y\|_p$$

Finally, we consider $|\cdot|$ -homogeneity.

If $\alpha \in \mathbb{R}$, $x \in \mathcal{L}_p$, we have

$$\begin{aligned} \|\alpha \cdot x\|_p &= \left(\sum_{k=1}^{\infty} |\alpha x|^p \right)^{\frac{1}{p}} = \left(\sum_{k=1}^{\infty} |\alpha| \cdot |x|^p \right)^{\frac{1}{p}} \\ &= |\alpha| \left(\sum_{k=1}^{\infty} |x|^p \right)^{\frac{1}{p}} = |\alpha| \cdot \|x\|_p. \end{aligned}$$

Space of bounded sequences

$$\ell_\infty = \left\{ x = (x_k)_{k=1}^\infty \in \mathbb{R}^\infty : \|x\|_\infty = \sup_{n \in \mathbb{N}} |x_n| < \infty \right\}$$

$(\ell_\infty, \|\cdot\|_\infty)$ is a normed space.

obvious: Non-negativity, non-degeneracy, 1-1 homogeneity.

Subadditivity is very similar to $(\mathbb{R}^n, \|\cdot\|_\infty)$

We have a distinguished subspace

$$c_0 = \left\{ x = (x_k)_{k=1}^\infty \in \ell_\infty : \lim_{k \rightarrow \infty} x_k = 0 \right\}$$

The vector space $C[a, b]$

let $a < b$ in $\mathbb{R}$.

$$C[a, b] = \left\{ f \in \mathbb{R}^{[a, b]} : f \text{ is cts} \right\}$$

$$\text{If } f, g \in C[a, b], \quad (f+g)(t) = f(t) + g(t)$$

$$\alpha f \in C[a, b]. \quad (\alpha f)(t) = \alpha \cdot f(t), \quad t \in [a, b]$$

Extreme Value Theorem

If $f \in C[a, b]$, then there are $t_{\min}, t_{\max} \in [a, b]$

for which $f(t_{\min}) \leq f(t) \leq f(t_{\max}), \forall t \in [a, b]$.

Hence define for $f \in C[a, b]$,

$$\|f\|_{\infty} = \max \{ f(t) \mid t \in [a, b] \} \quad (\exists \text{ by EVT})$$

we call it uniform norm.

And $(C[a, b], \|\cdot\|_{\infty})$ is a normed vector space.

Now for $p \in [1, \infty)$, define for $f \in C[a, b]$,

$$\|f\|_p = \left(\int_a^b |f(t)|^p \right)^{\frac{1}{p}}$$

($\exists$ b/c $|f(x)|^p$ is cts)

Thm : $(C[a,b], \|\cdot\|_p)$ is a normed space.

proof: Non-degeneracy, non-negativity, and 1.1 homogeneity of $\|\cdot\|_p$ are easy to check.

check subadditivity. If $f \in C[a,b]$, then notice that

$$\begin{aligned}\|f\|_p &= \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n \left| f\left(a+k \cdot \frac{b-a}{n}\right) \right|^p \left(\frac{b-a}{n} \right) \right)^{1/p} \\ &= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \left| f\left(a+k \cdot \frac{b-a}{n}\right) \right|^p \right)^{1/p} \left(\frac{b-a}{n} \right)^{1/p}\end{aligned}$$

Then if $g \in C[a,b]$, we have

$$\begin{aligned}\|f+g\|_p &= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \left(\left| f\left(a+k \cdot \frac{b-a}{n}\right) \right|^p + \left| g\left(a+k \cdot \frac{b-a}{n}\right) \right|^p \right) \right)^{1/p} \left(\frac{b-a}{n} \right)^{1/p} \\ &\leq \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \left| f\left(a+k \cdot \frac{b-a}{n}\right) \right|^p + \sum_{k=1}^n \left| g\left(a+k \cdot \frac{b-a}{n}\right) \right|^p \right)^{1/p} \left(\frac{b-a}{n} \right)^{1/p} \\ &= \|f\|_p + \|g\|_p\end{aligned}$$

□.

Topology: let (X, d) be a metric space. Given $x_0 \in X$ and $r > 0$ in $\mathbb{R}$. Define

• open ball ; $B(x_0, r) = \{x \in X : d(x, x_0) < r\}$

• closed ball ; $B[x_0, r] = \{x \in X : d(x, x_0) \leq r\}$

Ex: If $a < b$ in $\mathbb{R}$, then $(a, b) = B(\frac{a+b}{2}, \frac{b-a}{2})$,

$$[a, b] = B[\frac{a+b}{2}, \frac{b-a}{2}].$$

Def'n: let $U, F \in \mathcal{P}(X)$ where X is a metric space.

Then • U is open if $\forall x \in U$, there a $\varepsilon > 0$

$$\text{s.t. } B(x, \varepsilon) \subseteq U.$$

• F is closed if $X \setminus F$ is open.

Prop'n: let (X, d) be a metric space, $x_0 \in X$ and $r > 0$.

Then (i) $B(x_0, r)$ is open.

(ii) $B[x_0, r]$ is closed.

proof: (i). let $x \in B(x_0, r)$ and set $\varepsilon = r - d(x, x_0) > 0$.

Then for $y \in B(x, \varepsilon)$,

$$\begin{aligned} d(y, x_0) &\leq d(x, x_0) + d(y, x) \\ &< d(x, x_0) + \varepsilon \\ &= d(x, x_0) + r - d(x, x_0) \\ &= r \end{aligned}$$

Thus, $y \in B(x_0, r) \Rightarrow B(x, \varepsilon) \subseteq B(x_0, r)$.

(ii). let $x \in X \setminus B[x_0, r]$, and set $\varepsilon = d(x, x_0) - r$

Then $y \in B(x, \varepsilon)$, we have

$$\begin{aligned} d(x, x_0) &\leq d(x, y) + d(y, x_0) \\ &\leq \varepsilon + d(y, x_0) \\ &= d(x, x_0) - r + d(y, x_0) \\ \Rightarrow d(y, x_0) &\geq r \Rightarrow y \in X \setminus B[x_0, r]. \end{aligned}$$

X -metric space: $U \subseteq X$ is open if for any $x \in U$,

there is an $\varepsilon > 0$ s.t. $B(x, \varepsilon) \subseteq U$.

$F \subseteq X$ is closed iff $X \setminus F$ is open.

Example: i) let (X, d) be a discrete metric space.

with trivial metric.

$$d(x, y) = \begin{cases} 1 & , \text{ if } x \neq y \\ 0 & , \text{ if } x = y \end{cases}$$

Then for $x \in X$ and $r > 0$,

$$B(x, r) = \begin{cases} \{x\} & , r \leq 1 \\ X & , r > 1 \end{cases} \quad B[x, r] = \begin{cases} \{x\} & , r \leq 1 \\ X & , r > 1 \end{cases}$$

Therefore, singleton sets are open.

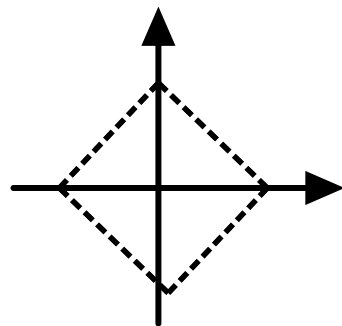
ii) Balls in $\mathbb{R}^n$: let for $p \in [1, +\infty]$.

$$B_p(0,1) = \{x \in \mathbb{R}^n : \|x\|_p < 1\}$$

Then $B_\infty(0,1) = (-1,1)^n$: unit cube

$$B_2(0,1) = \{x \in \mathbb{R}^n : x_1^2 + x_2^2 + \dots + x_n^2 < 1\} : \text{unit sphere.}$$

$$B_1(0,1) = \{x \in \mathbb{R}^n : x_1 + \dots + x_n < 1\} :$$



iii). If $f \in C[0,1]$, $r > 0$, then

$$B(f,r) = \{g \in C[0,1] : |f(t) - g(t)| < r, \forall t \in [0,1]\}$$

Remark: For $x_0 \in X$, metric space.

$$\text{We write } B[x_0, 0] = \{x_0\}$$

$\Rightarrow$ Singleton set is closed.

Prop'n : let (X, d) be a metric space.

i). Each of $\emptyset$, X is both closed and open.

ii). If $\{U_i\}_{i \in I}$ is a selection of open subsets of X .

then $U = \bigcup_{i \in I} U_i$ is open

iii). If $\{U_1, \dots, U_n\}$ is a finite selection of open subsets of X .

then $V = \bigcap_{i=1}^n U_i$ is open.

iv). If $\{\bar{F}_i\}_{i \in I}$ is a collection of closed sets,

then $\bigcap_{i \in I} \bar{F}_i$ is closed.

v). If $\{\bar{F}_1, \dots, \bar{F}_n\}$ is a finite collection of closed

subjects of X , then $\bigcup_{i=1}^n \bar{F}_i$ is closed.

proof :

Proof (i) Since for any $x \in X$ and any $\varepsilon > 0$,
 $B(x, \varepsilon) \subseteq X$. Hence X is open.
 Since for any $x \in \emptyset$ - of which there
 is none to choose - we have for
 any $\varepsilon > 0$ that $B(x, \varepsilon) \subseteq \emptyset$
 because there is nothing to test.
 Hence $\emptyset$ is open. Thus $X = X \setminus \emptyset$,
 $\emptyset = X \setminus X$ are closed.

-ii) If $x \in U$, then $x \in U_i$ for at least one i .
 Hence there is $\varepsilon > 0$ s.t.
 $B(x, \varepsilon) \subseteq U_i \subseteq U$.
 iii) If $x \in V$, $x \in U_i$ for each $i=1, \dots, n$
 so there $\varepsilon_i > 0$ s.t.
 $B(x, \varepsilon_i) \subseteq U_i$ for each i .
 Let $\varepsilon = \min \{ \varepsilon_1, \dots, \varepsilon_n \} > 0$
 Then $B(x, \varepsilon) \subseteq B(x, \varepsilon_i) \subseteq U_i$ for each i .
 iv), v) (DeMorgan's Law)
 $X \setminus \bigcap_{i \in I} F_i = \bigcup_{i \in I} (X \setminus F_i)$
 $X \setminus \bigcup_{i \in I} F_i = \bigcap_{i \in I} (X \setminus F_i)$
 are each open due to ii) and iii) above

Interior, Closure, Boundary, Desired Sets

let (X, d) be a metric space and $A \subseteq X$.

We define the interior by $A^\circ = \bigcup \{ U \in \mathcal{P}(X) \mid U \text{ is open, } U \subseteq A \}$

the closure by $\bar{A} = \bigcap \{ F \in \mathcal{P}(X) \mid F \text{ is closed, } F \supseteq A \}$

the boundary by $\partial A = \left\{ x \in X, \forall \varepsilon > 0, \begin{array}{l} B(x, \varepsilon) \cap A \neq \emptyset, \\ B(x, \varepsilon) \cap (X \setminus A) \neq \emptyset \end{array} \right\}$

the desired set by $A' = \{ x \in X, \forall \varepsilon > 0, (B(x, \varepsilon) \setminus \{x\}) \cap A \neq \emptyset \}$

Remark: i) $A^\circ \subseteq A$ is the largest open set in A .

$$A^\circ = A \Leftrightarrow A \text{ open}$$

ii) $A \subseteq \bar{A}$ is the smallest closed set containing A .

$$\bar{A} = A \Leftrightarrow A \text{ closed}$$

$$\text{iii) } \partial A = \partial (X \setminus A)$$

iv). we have $A \setminus A' = \{ x \in X, \text{ for some } \varepsilon > 0, B(x, \varepsilon) \cap A = \{x\} \}$

we call them isolated points of A .

Prop'n: For $A \subseteq X$ as above,

$$A^\circ = A \setminus \partial A$$

proof: let $x \in A$, then either

$$\textcircled{1} \exists \varepsilon > 0 \text{ s.t. } B(x, \varepsilon) \subseteq A$$

$$\Rightarrow B(x, \varepsilon) \cap (X \setminus A) = \emptyset. \text{ i.e., } x \notin \partial A.$$

$$\textcircled{2}. \forall \varepsilon > 0, B(x, \varepsilon) \not\subseteq A \Rightarrow B(x, \varepsilon) \cap (X \setminus A) \neq \emptyset.$$

$$\text{i.e., } x \in \partial A$$

Def'n: let $(x_n)_{n=1}^\infty \in X^{\mathbb{N}}$, $x_0 \in X$. We say that

" x_n converges to x_0 ",

written $\lim_{n \rightarrow \infty} x_n = x_0$ (or) $x_n \xrightarrow{n \rightarrow \infty} x_0$.

if $\forall \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}$ s.t. $n \geq n_\varepsilon \Rightarrow d(x_0, x_n) < \varepsilon$

Remark: If x_0 above exists, it's unique.

Indeed if also $\lim_{n \rightarrow \infty} x_n = y_0$, then

$$d(x_0, y_0) \leq d(x_0, x_n) + d(x_n, y_0) \xrightarrow{n \rightarrow \infty} 0$$

So $x_0 = y_0$. We call x_0 the limit of $(x_n)_{n=1}^{\infty}$.

Prop'n: For $A \subseteq X$ above,

$$A' = \left\{ x \in X : x = \lim_{n \rightarrow \infty} x_n \text{ for some } (x_n)_{n=1}^{\infty} \in (A \setminus \{x\})^{\mathbb{N}} \right\}$$

Furthermore, the sequence $(x_n)_{n=1}^{\infty}$ can be given to be
of distance points.

Proof: Let $x \in A'$. Then we find $x_1 \in (B(x, 1) \setminus \{x\}) \cap A$

and recursively find $x_{n+1} \in (B(x, \varepsilon_n) \setminus \{x\}) \cap A$,

$$\text{where } \varepsilon_n = \min \left\{ \frac{1}{n+1}, d(x, x_n) \right\}$$

Then $d(x, x_n) < d(x, x_{n-1}) < \dots < d(x, x_1)$

Showing that all points are distinct, and

$$d(x_n, x) < \frac{1}{n} \Rightarrow x_n \xrightarrow{n \rightarrow \infty} x$$

Conversely, if $x = \lim_{n \rightarrow \infty} x_n$, for some $(x_n)_{n=1}^{\infty} \in (A \setminus \{x\})^{\mathbb{N}}$.

then if $\varepsilon > 0$, we can find $n \in \mathbb{N}$ s.t. $d(x_n, x) < \varepsilon$

$$\text{i.e. } x_n \in (B(x, \varepsilon) \setminus \{x\}) \cap A, \Rightarrow x \in A'$$

is.

Thm (Description of closure)

The closure $\bar{A}$ may be given by any of the following description:

$$A \cup \partial A, \quad A \cup A'$$

$$(\text{ball-meet set}): A_m = \{x \in X : \forall \varepsilon > 0, B(x, \varepsilon) \cap A \neq \emptyset\}$$

$$(\text{limit point set}): A_L = \{x \in X : x = \lim_{n \rightarrow \infty} x_n, \text{ for some } (x_n)_{n=1}^{\infty}\}$$

proof:

Proof: We have

$$\begin{aligned}
 \overline{A} &= \bigcap \{F \in \mathcal{P}(X) : F \text{ is closed and } A \subseteq F\} \\
 &= \bigcap \{X \setminus U : U \text{ is open in } X \text{ and } U \subseteq X \setminus A\} \\
 &= X \setminus \bigcup \{U \in \mathcal{P}(X) : U \text{ is open and } U \subseteq X \setminus A\} \\
 &= X \setminus [X \setminus A]^{\circ} \quad (\text{def'n of interior}) \\
 &= X \setminus ((X \setminus A) \setminus \partial(X \setminus A)) \\
 &= X \setminus ((X \setminus A) \setminus \partial A) = A \cup \partial A.
 \end{aligned}$$

If $x \in A \cup \partial A$, then for every $\varepsilon > 0$,
 $B(x, \varepsilon) \cap A \neq \emptyset$ so $x \in A_M$.
 Conversely if $x \in A_M$, then either
 for every $\varepsilon > 0$, we have
 $B(x, \varepsilon) \cap (X \setminus A) \neq \emptyset$, so $x \in \partial A$,
 (or) for some $\varepsilon > 0$, we have
 $B(x, \varepsilon) \subseteq A$, so $x \in A^{\circ}$.
 Thus $x \in A_M$ implies that $x \in \partial A \cup A^{\circ} \subseteq \partial A \cup A$
 $\therefore A \cup \partial A = A_M$.

Similarly, we see that $A \cup A' = A_M$.

It's an immediate application of the definition of a

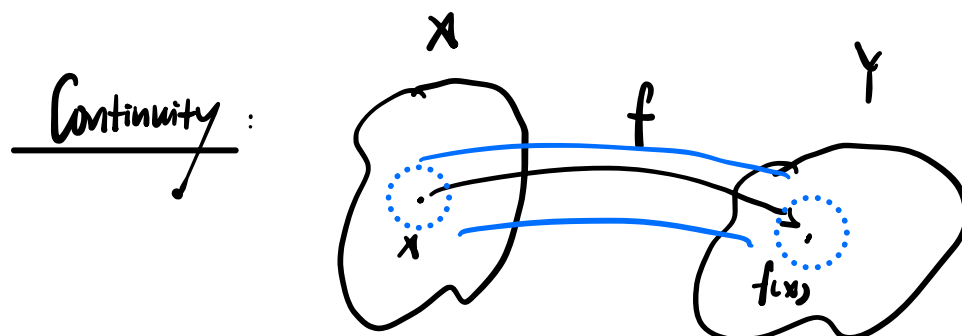
limit that $A_M = A_L$

□.

Cor: A is closed $\Leftrightarrow \partial A \subseteq A \Leftrightarrow A' \subseteq A$

$\Leftrightarrow A = A_M$

proof: as A is closed $\Leftrightarrow A = \bar{A}$



Def'n: let (X, d_X) , (Y, d_Y) be metric spaces.

$f: X \rightarrow Y$ a function, we say that

f is continuous at $x_0 \in X$ if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } d_X(x, x_0) < \delta$$

$$\Rightarrow d_Y(f(x), f(x_0)) < \varepsilon \text{ in } Y \quad (*)$$

We say f is cts on X if f is cts

at each point in X .

$$(*) \Leftrightarrow f(B(x_0, \delta)) \subseteq B(f(x_0), \varepsilon)$$

$$\Leftrightarrow B(x_0, \delta) \subseteq f^{-1}(B(f(x_0), \varepsilon))$$

We call $N \subseteq X$ a nbh. of x_0 if $x_0 \in N^\circ$ (interior)

Thm: (Continuity at a point)

Given metric space (X, d_X) , (Y, d_Y) , $x_0 \in X$.

and $f: X \rightarrow Y$. TFAE.

- (i). f is cts at x_0
- (ii). for every nbh N of $f(x_0)$, $f^{-1}(N)$ is a nbh of x_0 .
- (iii). if $\lim_{n \rightarrow \infty} x_n = x_0$ in X , then $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ in Y .

proof: (i) $\Rightarrow$ (ii). By assumption on N , $\exists \varepsilon > 0$ s.t. $B(f(x_0), \varepsilon) \subseteq N$.

Since f is cts, $\exists \delta > 0$ s.t. $B(x_0, \delta) \subseteq f^{-1}(B(f(x_0), \varepsilon)) \subseteq f^{-1}(N)$

$\Rightarrow f^{-1}(N)$ is n.b.h. of x_0 .

(ii) $\Rightarrow$ (i). Given $\varepsilon > 0$, $B(f(x_0), \varepsilon)$ is a nbh of $f(x_0)$

So $f^{-1}(B(f(x_0), \varepsilon))$ is a nbh of x_0 ,

which provides for some $\delta > 0$ s.t. $B(x_0, \delta) \subseteq f^{-1}(B(f(x_0), \varepsilon))$ (+)

(i) $\Rightarrow$ (iii). Assume (†). If $\lim_{n \rightarrow \infty} x_n = x_0$, then $\exists n_\varepsilon \in \mathbb{N}$

s.t. $\forall n \geq n_\varepsilon$, we have $x_n \in B(x_0, \delta) \subseteq f^{-1}(B(f(x_0), \varepsilon))$

$\Rightarrow f(x_n) \in B(f(x_0), \varepsilon)$.

$\Rightarrow \lim_{n \rightarrow \infty} f(x_n) = f(x_0) \text{ in } Y$.

(iii) $\Rightarrow$ (i). (Contrapositive): If (i) fails, $\exists \varepsilon > 0$ s.t. $\forall \delta > 0$,

we have $B(x_0, \delta) \not\subseteq f^{-1}(B(f(x_0), \varepsilon))$

Hence for each $n \in \mathbb{N}$, we may find

$$x_n \in B(x_0, \frac{1}{n}) \setminus f^{-1}(B(f(x_0), \varepsilon))$$

Thus $d(x_n, x_0) < \frac{1}{n}$, so $x_n \xrightarrow{n \rightarrow \infty} x_0$, but $f(x_n) \notin B(f(x_0), \varepsilon)$

$\#$.